Ch 9.1-2: Maximum Margin Classifier and Support Vector Classifier

Lecture 23 - CMSE 381

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Mon, Apr 11, 2022

Announcements

Last time:

- Ch 8: Decision Trees

This lecture:

- Maximal Margin Classifier
- Support Vector Classifier

Announcements:

- Mon 4/11: 9.1-2
- Weds 4/13: 9.3-4
- Fri 4/1: Lab: 9.6
  - HW #9 Due Friday

Section 1

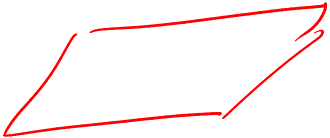
Maximal Margin Classifier

What is a hyperplane?

In p-dimensional space, hyperplane is (p-1)-dimensional affine subspace.

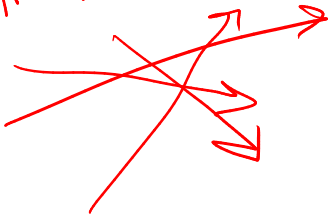
ex) $p=3$

hyperplane 2-dim plane



$p=2$

hyperplane is line



Mathematical definition of a hyperplane

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p = 0$$

coeff determine hyperplane

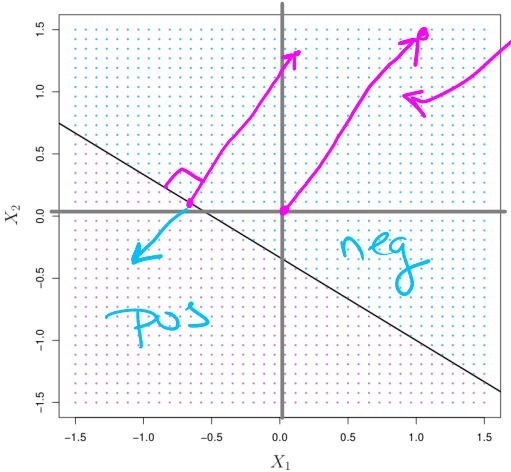
Pt on the hyperplane $(x_1, \ldots, x_p)$ is anything that satisfies above equation

$p = 2$ $\quad \beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0 \quad \beta_0 + \beta_1 x + \beta_2 y = 0$

int slope

$$\frac{-\beta_0}{\beta_2} - \frac{\beta_1}{\beta_2} x = y = \frac{-\beta_0 - \beta_1 x}{\beta_2}$$

Hyperplane for $p = 2$



$$1 + 2X_1 + 3X_2 = 0$$

Normal $(2,3)$

$(1, 1.5)$

Unit normal:

Same direction, unit length

$(2,3)$ has length $\sqrt{4+9} = \sqrt{13}$

$$\left(\frac{2}{\sqrt{13}}, \frac{3}{\sqrt{13}}\right)$$

There are two sides to every hyperplane

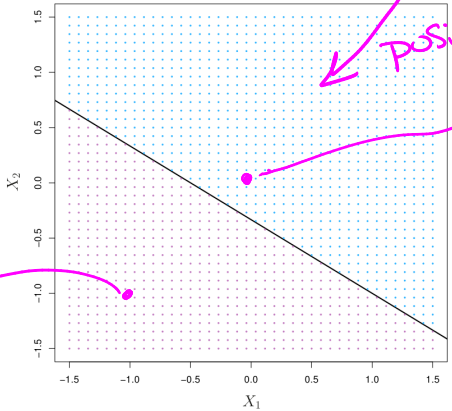
$$\underbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p < 0}$$

$$\underbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p > 0}$$

positive

pt (0,0)

$1+2(0)+3(0)=1$



$(-1,-1)$

$1+2(-1)+3(-1)$

$=1-2-3=-4$

$1+2x_1+3x_2=0$

Classification Setup

Data matrix:

$$X = \begin{pmatrix} - & x_1^T & - \\ - & x_2^T & - \\ & \vdots & \\ - & x_n^T & - \end{pmatrix}_{n \times p}$$

$$x_1 = \begin{pmatrix} x_{11} \\ \vdots \\ x_{1p} \end{pmatrix}, \cdots, x_n = \begin{pmatrix} x_{n1} \\ \vdots \\ x_{np} \end{pmatrix}$$

Observations in one of two classes,
$y_i \in \{-1, 1\}$

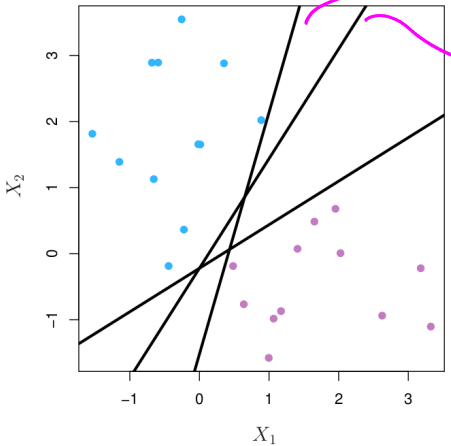
$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

Separate out a test observation

$$x^* = (x_1^* \ \cdots \ x_p^*)^T$$

Separating Hyperplane

$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} > 0 \text{ if } y_i = 1$$
$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} < 0 \text{ if } y_i = -1$$



Separating hyperplane

↳ all points of type 1 are on one side and all of type -1 on the other side

Strict - No errors allowed

Another way to say it

$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} > 0 \text{ if } y_i = 1$$
$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} < 0 \text{ if } y_i = -1$$

If $y_i = -1$

$f(x_i) \overset{<}{\cancel{>}} 0$

$y_i = 1$

For all i:

$$y_i \underbrace{(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip})}_{f(x_i)} > 0$$

True ✓

Separating hyperplane becomes a classifier

If you have a separating hyperplane:

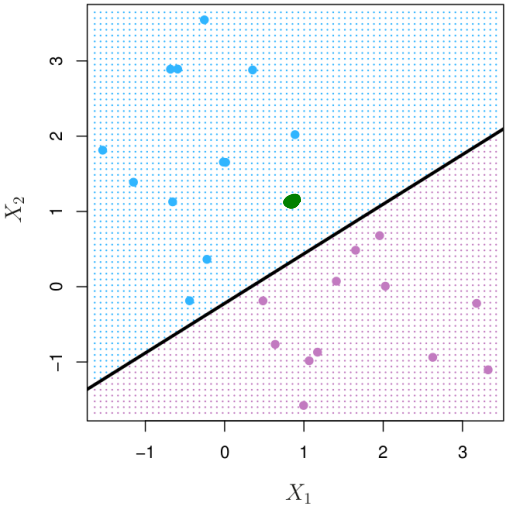
- Check

$$f(x^*) = \underbrace{\beta_0 + \beta_1 x_1^* + \beta_2 x_2^* + \cdots + \beta_p x_p^*}$$

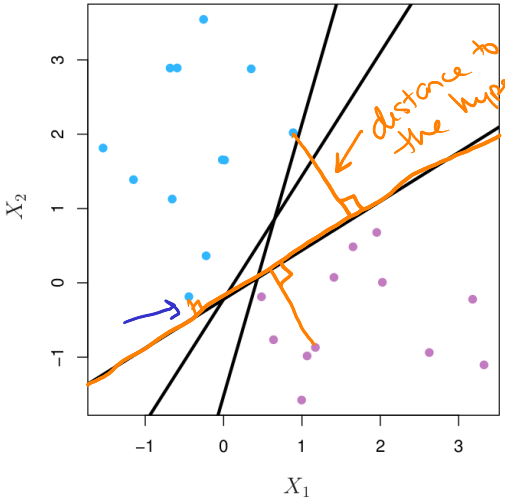
- If positive, assign $\hat{y} = 1$
- If negative, assign $\hat{y} = -1$

$x^* = (1,1)$

$f(x^*) = \beta_0 + \beta_1(1) + \beta_2(1)$



How do we pick?

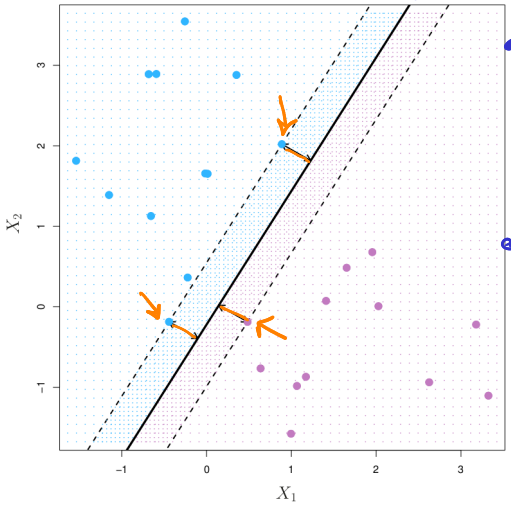


Lots of separating hyperplanes (if there's at least 1)

Which is best?

The margin is the smallest distance from any data point to the hyperplane

Maximal margin classifier



- Maximal margin hyperplane is the separating hyperplane with the largest margin
- Use it to get maximal margin classifier
- Observations that are closest are called support vectors

Only effect on MMH is by location of support vectors

Mathematical Formulation

$$\underbrace{\underset{\beta_0,\beta_1,\ldots,\beta_p,M}{\text{maximize}}}\ M$$

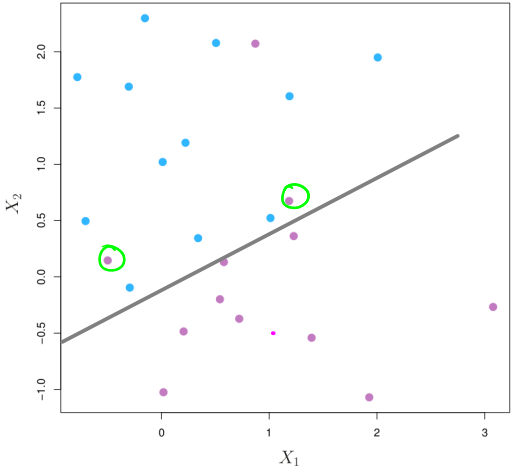
margin

$$\text{subject to} \sum_{j=1}^{p} \beta_j^2 = 1,$$

forces normal vector to be unit length

$$\underbrace{y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip})}_{\text{distance to hyper plane from point } x_i = (x_{i1}, x_{i2}, \ldots, x_{ip})} \geq M \quad \forall\, i = 1, \ldots, n$$

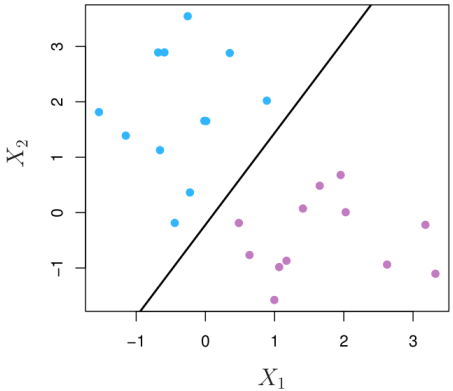
But what if....

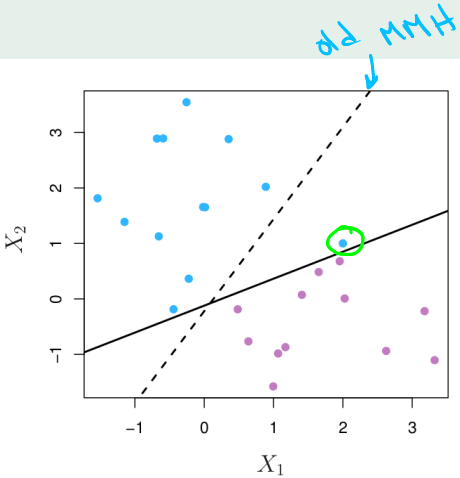


.... there is no separating hyperplane

No solution to above optimization

Sensitivity to new points





Even if you have seperating hyper plane, it might not be good

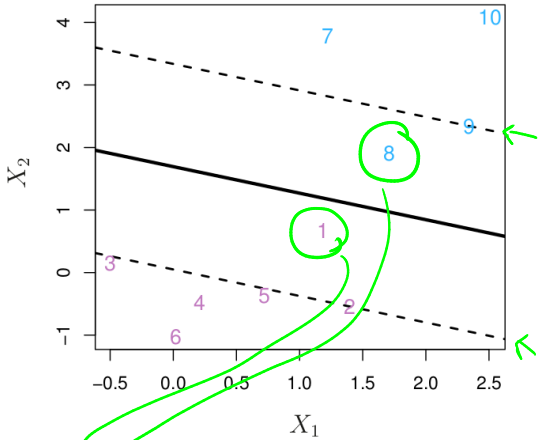
Section 2

Support Vector Classifier

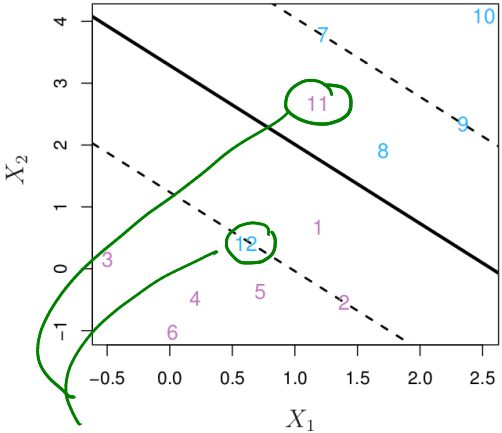
Basic idea

- Be ok with having a classifier that's not quite perfect
- Aim for
  - robust to individual observations
  - Better classifier for <u>most</u> of data
- Soft margin classifier

Soft margin



Correct side of hyperplane
Wrong side of margin



misclassified
Wrong side of hyperplane

Math version of SVC

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

width of the margin

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \epsilon_i),$$

ε_i are slack variables allowing observations to go to the wrong side

$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C,$$

non-negative tuning parameter

What is ε?

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

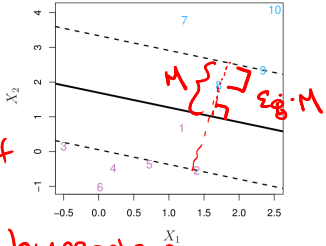
$$\underbrace{y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \epsilon_i)}_{\text{distance to hyperplane}},$$

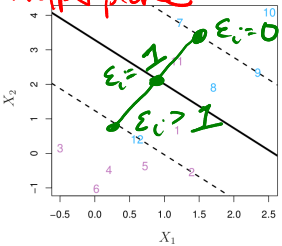
$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C,$$

If $\varepsilon_i = 0$, x_i is on correct side of margin

If $\varepsilon_i > 0$ wrong side of margin

If $\varepsilon_i > 1$ wrong side of hyperplane





What is C?

$C=0 \Rightarrow$ no pts on wrong side of margin

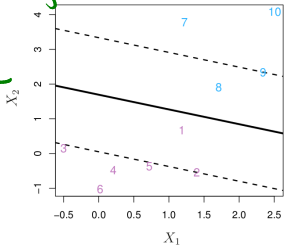
$\Rightarrow$ max'l margin hyperplane

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

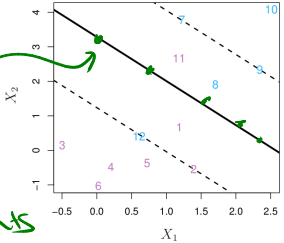
$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1-\epsilon_i),$$

$$\epsilon_i \geq 0, \quad \underbrace{\sum_{i=1}^{n} \epsilon_i \leq C,}$$



$\epsilon_i = 1$

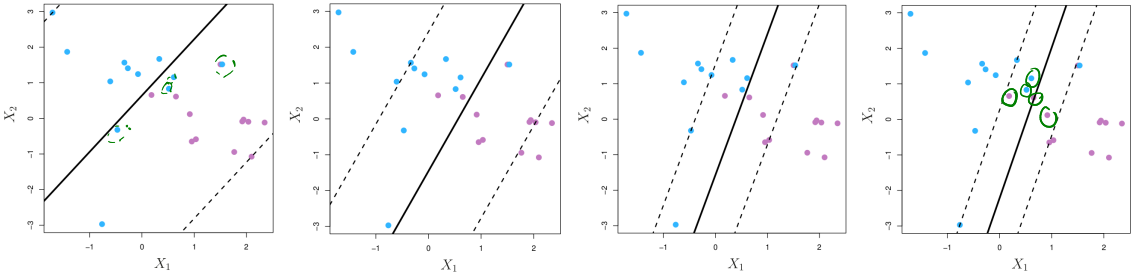


All $\epsilon_i = 1 \Rightarrow$ Sum is a const

At most C points can violate the margin

Bounds ϵ_i's, controls number + severity of violating margin

Examples messing with C



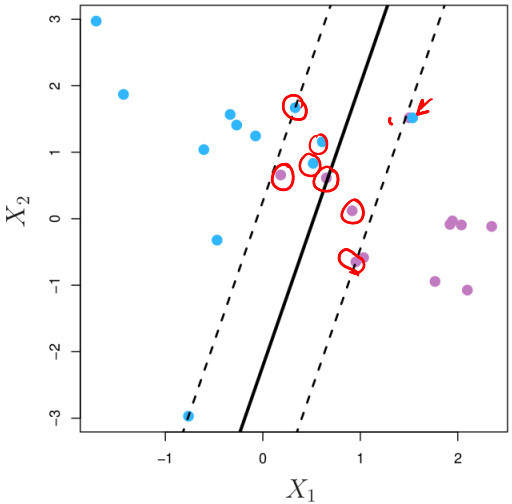
Large C

wide margin

Decreasing $C \rightarrow$

narrow margin

What affects the hyperplane?



Only the observations on or on the wrong side of the margin affect the hyperplane

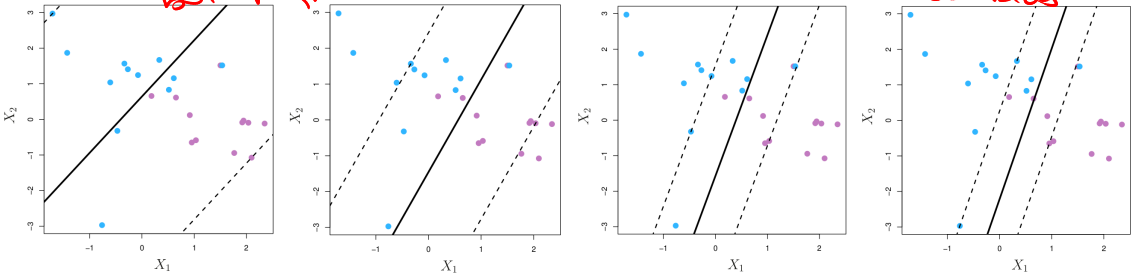
These are called support vectors

Changing other points' positions do not affect hyperplane

Bias-variance tradeoff

Big C
Big margin
Lots of support vectors
⇒ Low variance but high bias

Small C
Small margin
Few support vectors
⇒ High variance but low bias



Decreasing C →

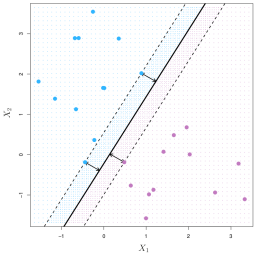
TL;DR

Maximal Margin Classifier

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,M}{\text{maximize}} \; M$$

$$\text{subject to} \; \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M \;\; \forall\, i = 1, \ldots, n$$



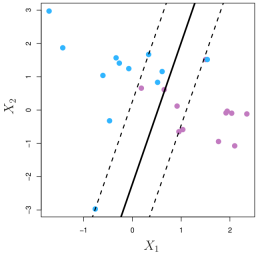
Support Vector Classifier

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\, M}{\text{maximize}} \quad M$$

$$\text{subject to} \; \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \epsilon_i),$$

$$\epsilon_i \geq 0, \;\; \sum_{i=1}^{n} \epsilon_i \leq C,$$



Everything has a linear boundary

Next time

Next time:

- 9.3/9.4

This week:

- Mon 4/11: 9.1-2
- Weds 4/13: 9.3-4
- Fri 4/1: Lab: 9.6
  - HW #9 Due Friday

A touch short
more examples would
help